

Total No. of Questions : 8]

SEAT No. :

PE2733

[Total No. of Pages : 2

[6583]-314

T.E. (E & TC Engineering) (Honors)
ROBOT PROGRAMMING & SIMULATION
(2019 Pattern) (Semester - VI) (304183 HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) What is VAL Language in Robotics? [6]
b) Explain palletizing applications using VAL. [6]
c) Explain VAL Program for PICK and PLACE robotic applications. [6]

OR

- Q2)** a) Explain Robot welding application using VAL program. [6]
b) What are the motion commands used in robotics? [6]
c) Explain Robot welding application using VAL program-WAIT, SIGNAL and DELAY command for communications using simple applications. [6]

- Q3)** a) What is rapid language for Robotics? What are advantages of Robotics RAPID language? [6]
b) What are instructions in automatic mode and subroutine command based programming for PICK and PLACE robotics? [6]
c) What is Motion, Sensor commands-Data processing? [5]

OR

- Q4)** a) What are Motion Instructions-Pick and place operation using Industrial robot-manual mode? [6]
b) What is the Difference Between Robotics and Automation. [6]
c) What are elements and functions, Statements, constants and variables in AML? [5]

P.T.O.

- Q5)** a) What is soft robotics? [6]
b) How AR and VR used in Robotics? [6]
c) Explain Robot studio online software. [5]

OR

- Q6)** a) Explain concept of RPA. [5]
b) What the terms resolution, accuracy and repeatability are as applied to robotics? [6]
c) What is jogging in a robot? What are the 5 main components of robots? [6]

- Q7)** a) Explain Monte Carlo Localization for Mobile Robot. [9]
b) What are the 4 types of models in simulation in robotics? Why is the Cobweb theory important? [9]

OR

- Q8)** a) What is the continuous cobweb model? [9]
b) Explain software used for design robotics application. [9]

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